SLIIT
Discover Your Future

BSc (Hons) in Information Technology Specializing Data Science

Research Project - IT4010 Ethical Analysis Report

Group ID: R26-DS-010

Project Title: Multimodal Computational Intelligence for Improving Caregiver Performance and Care Delivery

IT Number: IT22057488

Individual Component Title: Edge-Enabled Skeletal Movement Analysis System for Real-Time Patient Observable Fall Risk Detection and Prediction

	Identification and Analysis
Data Privacy Concerns	SENTRY uses RGB-camera input to extract skeletal joint coordinates for fall-risk analysis. The system was designed to reduce unnecessary exposure of identifiable visual data, but skeleton data are still treated as personal data because they may be linked with movement patterns, timestamps, room information or patient codes. **Key Measures:** • Raw video is not stored during live operation. • Faces are blurred in the dashboard stream • Alert replay data contains skeleton coordinates rather than raw video • Uploaded videos used for offline analysis are deleted after skeleton extraction. • Original volunteer recordings are retained locally for research verification and are not uploaded to cloud services. • Both volunteers gave consent before recording. • Future deployment would require access controls, retention rules and appropriate handling of personal data.
Data Security	SENTRY uses cloud-backed services for operational data and replay storage, so security of credentials, access and communication is important. The current system is a development prototype and has not yet been hardened for real clinical use **Key measures and concerns:** • Supabase Row-Level Security is enabled for SENTRY database tables. • Credentials are stored in environment variables rather than hard coded in source code. • Optional skeleton replay files can be stored in Cloudflare R2. • The current API and WebSocket connections do not implement authentication. • CORS is currently open and the backend uses a Supabase service-role key. • Database outages may cause events to be dropped. • Real deployment would require authentication, role-based access control, encrypted transport, restricted CORS, secure credential management and a data-retention policy.

Bias and Fairness	The available datasets do not represent all possible patient groups or care environments. Most public training data contain staged actions by healthy adults, and the in-house webcam dataset is very small, so the current system may perform differently across people and settings. **Main bias risks:** • The webcam dataset contains only two volunteers. • SENTRY has not been tested on frail older adults, wheelchair users, people using walking aids, bedridden patients or people with different mobility conditions. • Camera angle, lighting, body shape and movement style may affect pose estimation. • Cross-dataset testing showed that performance can drop when the sensor or environment changes. • Results should not be assumed to generalize equally to all patient groups. • Future work should use larger and more diverse datasets and separate evaluations across different user and environmental conditions.
Transparency and Accountability	The thesis documents how SENTRY works, how it was trained, how it was evaluated and where its limitations remain. The system is intended as a caregiver-support tool rather than a replacement for professional judgement. **Transparency and accountability measures:** • The datasets, preprocessing, model architecture, biomechanical features, training process and evaluation methods are documented. • SENTRY reports NORMAL, MODERATE and HIGH risk states. • The interface can show unusual biomechanical factors, but these are descriptive and not a full explanation of the neural-network decision • The system does not claim to forecast falls several seconds or minutes in advance • The prototype has not been clinically validated and is not claimed to be a certified medical device. • Future deployment should include audit logs, alert-review procedures and clearly defined responsibility for false or missed alerts.
Impact on Stakeholders	SENTRY may provide useful support to caregivers by identifying fall-like movement and generating alerts, but incorrect output could also affect patients and staff. The main stakeholders are patients, caregivers, nursing staff, ward managers and care facilities. **Possible impacts:** • Earlier awareness of fall-like movement may help staff respond more quickly.

	• A missed fall could delay assistance or create false reassurance • Repeated false alerts could increase workload and contribute to alarm fatigue. • Camera-based monitoring may raise privacy concerns even when faces are blurred. • Caregivers need to understand that the system provides decision support and not a clinical diagnosis. • The system has not yet established clinical effectiveness or acceptable real-world alert rates. • Future testing should involve healthcare stakeholders and assess safety, privacy, usability and alert burden before deployment.